

KATIA SIVKOVA

PhD student at LIRA [↗](#), visiting student at FCLA IRL 3386 [↗](#)

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EXPERIENCE

PhD student

[Observatoire de Paris](#) [↗](#)

[📅](#) September 2025 – Ongoing [📍](#) Paris/Santiago

High accuracy distances of Cepheids from optical interferometry and Gaia DR4

- Currently working on simulating DR4 epoch astrometry of variable-induced movers
- Working on GRAVITY data

Master thesis

[Observatoire de la Côte d'Azur](#) [↗](#)

[📅](#) March 2025 – August 2025 [📍](#) Nice, France

Visible and near-infrared observations of fast-rotators with the new generation of instruments at the CHARA array

- Worked on implementing custom priors for MCMC within the oimodeler Python package [↗](#)
- Worked on fitting MIRC-X & MYSTIC & SPICA/CHARA data for α Cep, α Oph, and α Lyr to constrain gravity darkening

Internship

[Centro de Astrobiología \(CAB\)](#) [↗](#)

[📅](#) January 2025 – March 2025 [📍](#) Madrid, Spain

Identifying binaries in a cluster using VO tools

- Worked on detecting binary systems in the IC 348 and NGC 1333 clusters using binary SED fitting of VOSA

Internship

[ACRI-ST](#) [↗](#)

[📅](#) December 2023 – January 2024 [📍](#) Grasse, France

Data fusion of remote sensing data: Hyperparameters tuning with Bayes

- Worked on hyperparameter optimization for a CNN developed for Sentinel data fusion within the Copernicus mission

Bachelor thesis

[Observatoire Midi-Pyrénées](#) [↗](#)

[📅](#) February 2023 – May 2023 [📍](#) Tarbes, France

Measurement of rotational velocity and impact of pulsation/anisotropy of brightness

- Worked on converting a spectral line profile simulation code for rotating stars from Fortran to Python
- Worked on developing a code to measure stellar rotation using the Fourier method and applied it to spectra from Neo-Narval/OMP and ESPaDOnS/CFHT

LANGUAGES

English (C1) French (C2) Russian (Native)

EDUCATION

Ph.D. in Astrophysics [↗](#)

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[📅](#) Sept 2025 – Present

M.Sc. in Astrophysics [↗](#)

[Université Côte d'Azur](#)

[📅](#) Sept 2023 – June 2025

B.Sc. Physics, Chemistry, Astrophysics, Meteorology and Energy [↗](#)

[Université Toulouse III Paul Sabatier](#)

[📅](#) Sept 2021 – May 2023

1st year of B.Sc. in Physics, Chemistry [↗](#)

[Université de Pau et des Pays de l'Adour](#)

[📅](#) Sept 2020 – May 2021

PUBLICATIONS

[📄](#) Refereed Articles

- N. Nardetto et al., "Simultaneous optical and infrared high spectral resolution observations of cepheids," *submitted to A&A*,
- K. Sivkova et al., "Effect of binarity and variability on the astrometry of unresolved systems," *in prep. for A&A* [↗](#),
- A. Gallenne et al., "High-contrast imaging of galactic cepheids with vlt/sphere," *accepted to A&A* [↗](#), Mar. 2026.
- V. Hocdé et al., "Reconstruction of cepheid radial velocity curves from the shape of the v-band light curves," *accepted to A&A* [↗](#), Mar. 2026.

[👥](#) Conferences

- K. Sivkova, "Distances of binary Cepheids from combined Gaia DR4 astrometry and optical interferometry," in *CHARA Science Meeting 2026*, talk, Mar. 2026.
- K. Sivkova, "Preparation for Gaia DR4: Simulations of epoch astrometry for non-single stars," in *ExoSystèmes V*, talk, Mar. 2026.
- P. Bendjoya et al., "The calern asteroid polarimetric survey version 2.0," in *EU-ESA Workshop on Size Determination of Potentially Hazardous Near-Earth Objects* [↗](#), 2024.